Consistency, reversibility, and the limits of causal-loop explanation

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Abstract

A closed loop can be logically consistent, in a precise operational sense, without having a fixed global causal order, and that consistency can be exactly what one reversible extension requires. Finitely many parties choose functions from their own inputs to their own outputs. A process function sends the outputs back to the inputs so that every such choice has exactly one history. Some of these functions are cyclic: each party's input depends on another party's output, and no party is first. For finite alphabets, the canonical reversible extension of a process function stays unitary when arbitrary local unitaries and finite ancillas are inserted, and it does so for every such insertion if and only if the classical rule is a process function. Consistency and this reversible realization therefore coincide. Neither is a fixed causal order, and neither is a license for physical or metaphysical time travel. A further identification fails. Mixtures of process functions arise from pure quantum processes, but purity does not return only those mixtures. An explicit three-party process, unitary under every local unitary, induces in the computational basis a valid classical distribution outside the convex hull of the process functions. The distribution is tied to that basis, and the underlying board is a known one read in a stated frame. The argument separates constraints that the time-travel dialectic and the recent literature on causal order sometimes bundle together. It does not refute those views.

Keywords. causal loops; logical consistency; indefinite causal order; process functions; reversibility; purification

1 Introduction

Stories about time travel are often asked to do two jobs at once. They are asked to be consistent, and they are asked to be stories about a world in which someone really travels. The first job has a precise form. If a later action is among the causes of an earlier condition, the two have to fit. The grandfather lives long enough for the traveler to exist, or the shot is never the shot that would have prevented that existence. The second job is different. It asks whether any such fit is a physical process, a metaphysical possibility, or a freedom the traveler can exercise. Much of the philosophical discussion treats a positive answer to the first question as the main obstacle to the second, or treats a remaining mystery about the first as a reason to doubt the second (Lewis, 1976; Loewenstein, 2022).

A different literature, in the foundations of quantum theory, has built models in which several laboratories exchange systems with no assumption that the laboratories sit in one causal order (Oreshkov et al., 2012; Baumeler and Wolf, 2016). The deterministic classical points of that theory are process functions. Each party may apply any local function. The global rule is admitted only when every such choice has exactly one consistent history. Baumeler et al. (2019) show that deterministic classical dynamics of this kind, including arrangements in which three parties are each in the past and the future of the others, can be realized by reversible interactions. Adlam (2023) argues that the resulting notion

of causal order is real enough to matter in fundamental physics and weaker than ordinary causation, because it does not require asymmetry. Araújo et al. (2017) propose that a process is a candidate for being physical only when it is part of a pure process: whenever the parties apply unitaries, the whole process induces a unitary from the past to the future.

These models are easy to misread as answers to the time-travel question. A consistent loop looks like a reply to the grandfather paradox. A reversible realization looks like a physical mechanism. A pure quantum process looks like a deeper story of which the classical loop is only the surface. The argument of this article is that those readings attach the models to the wrong conclusions. Universal consistency under a specified class of local interventions can coexist with the failure of any fixed global causal order. For finite alphabets, that consistency condition is necessary and sufficient for one canonical reversible extension to remain unitary under every local unitary. Neither fact is a spacetime theorem, and neither is a theorem about metaphysical possibility. Purification does not restore the missing identification. A pure process can induce, in a chosen classical basis, statistics that no mixture of deterministic consistent processes yields.

The mathematical evidence is three results from an unpublished technical note by Arrowood.[1] First, every process function on finite alphabets has a canonical reversible extension that remains unitary for every choice of local unitaries and finite ancillas. Second, within that same family of tests, the closed loop is unitary for every such choice only if the classical rule is a process function. The second result is restricted. It is not the claim that unitarity and consistency coincide for every dilation anyone might write down. Third, there is a three-party binary process, built from a known board and read in the computational basis, that is valid and purifiable and that lies outside the convex hull of the process functions. An earlier conjecture, that the purifiable classical processes were exactly the mixtures of process functions, is false in that case. It is not thereby settled which other classical processes are purifiable.

The article is organized as follows. Section 2 says what the consistency condition quantifies over, and separates deterministic histories from probabilistic descriptions. Section 3 says what a fixed causal order would add, and why the three-party examples do not have one. Section 4 explains the reversible extension and the sense in which it fails to be a time-travel license. Section 5 engages Lewis and Loewenstein: the condition is a candidate for what "does logic's bidding," and an honest account of why that candidate does not answer the inexplicability objection. Section 6 turns to purification, the chosen basis, and Adlam's distinction between causal order and ordinary causation. Section 7 answers the objections that the results are philosophically idle, that the constraint is trivial, that the basis spoils the separation, and that the infinite-party boundary is a set-theoretic artefact. Section 8 states the conclusion and the open edges. The appendix records the definitions and the short proofs. A note on assistance and on declarations that are still pending is Section 9.

Two restrictions are in force throughout. The parties and the alphabets in the theorems are finite. Where an infinite-party statement is mentioned, it is a boundary of the finite theory, and the general question about exclusive maps of Baire class one is left open. Nothing below is offered as a survey of time-travel metaphysics or of quantum causal modelling. The views that are discussed are discussed because the three results change what those views can treat as the same constraint.

[1]The note is unpublished. A version prepared so that reviewers can inspect the proofs without the author's name has not been made, and this article does not pretend that such a version exists. The appendix states the definitions and the short arguments on which the philosophy depends. Machine-checked certificates are part of the unpublished note; they certify formal statements, not the readings given here.

2 Consistency under local interventions

2.1 Deterministic histories

Consider finitely many parties. Each has a finite set of possible inputs and a finite set of possible outputs. A local intervention, in the deterministic sense used here, is a function for each party from that party's input alphabet to that party's output alphabet. The function may be any function of that type. It is not required to be a function of a background time, and it is not required to factor through a designated cause. The parties' functions are chosen independently. A global rule w then sends a tuple of outputs to a tuple of inputs. Composing the local functions with w produces a function from output tuples to output tuples. A fixed point of that function is a history: a tuple of outputs that, fed through w and then through the local functions, yields itself.

The consistency condition is universal and exact. For every choice of local functions, there is exactly one fixed point. A rule that meets the condition is a process function (Baumeler and Wolf, 2016; Adlam, 2023). Zero fixed points is a contradiction. The local choice asks for a history, and the rule supplies none. Two or more fixed points is a different failure. The local choice does not determine which history occurs. Both failures are excluded by the same clause. The clause does not say that the parties are forbidden some local function. It says that a rule which cannot live with every local function is not admitted.

This is a specific intervention. It is not an intervention in the sense of a surgical change to a variable inside a causal graph, and it is not an intervention by an agent whose freedom has been given a metaphysical analysis. It is the act of inserting a function of a stated type. Stochastic local maps come later, as mixtures of these functions. Unitary local maps come later still, as a different test of the same rule. Keeping the three tests apart is the point of the next several sections. A probabilistic description is not a deterministic history that we have failed to inspect. A unitary insertion is not a classical function written in quantum notation.

The quantifier over local functions is what makes the condition strong. It is easy to write a rule that has one history for the interventions one happens to consider. The grandfather story, as usually told, fixes one intention and one occasion. A process function is answerable to every function on the alphabets, including the functions that flip a bit, copy a bit, ignore the input, or encode a complicated dependence on it. Adlam's one-laboratory example is the simplest casualty. If a channel around a single laboratory is the identity, and the laboratory applies a bit flip, the output is required to equal its own negation (Adlam, 2023). That pair has no history. The identity channel is not a process function (Adlam, 2023). The exclusion is not an extra ban on bit flips. The bit flip remains available. The channel is what fails.

2.2 Probabilistic descriptions

A process function is deterministic. It returns an input tuple, not a distribution. The classical theory around it also contains probabilistic processes. A classical process, in the sense needed below, assigns a probability to each input tuple given each output tuple, and it assigns total probability one under every product of local stochastic maps. The deterministic local functions are the extreme points of those maps. Because the total-probability condition is multilinear, a process that is normalized on every tuple of deterministic functions is normalized on every tuple of stochastic maps. Validity in this sense is weaker than being a process function. A process function induces a valid process that is concentrated on its unique history. Many valid processes are not concentrated on a single history.

Mixtures sit between the two. If several process functions are combined with weights, the result is a valid classical process whose randomness can be read as ignorance of which deterministic rule is in force. The convex hull of the process functions is the set of all such mixtures. Membership in the hull is

a probabilistic description with a special structure: every component is a deterministic rule that itself meets the universal fixed-point condition. A valid process outside the hull cannot be given that reading. Its probabilities are not the probabilities of a hidden process function.

The distinction matters for the later purification claim, and it already matters for the grandfather story. "Something prevents the killing" can mean that, in the one history that occurs, a banana peel or a misfire occurs. It can also mean that the only rules we are willing to count as possible are rules for which every intention, including the intention to kill, has exactly one non-killing history. Those are different claims. The first is about an event in a history. The second is about which rules are admitted before any history is selected. A probabilistic mixture of process functions does not blur the difference. It says that one of several fully consistent deterministic rules is in force, and it does not say which.

The difference can be seen on the same three bits. Suppose two process functions are mixed with equal weights. For any local intervention, each component contributes exactly one history, and the mixture assigns probability one half to each of those histories, or probability one if they coincide. The randomness is ignorance of the component. Nothing in the mixture assigns positive probability to a pair of outputs that no process function can produce together with the observed inputs. The witness of Section 6.2 fails exactly that test. Its conditional probabilities, in the computational basis, give a negative value to a linear functional that is nonnegative on every process function. No weights on the 744 deterministic rules reproduce it. A story in which "something or other stops the killing" can still be told inside the hull: each component stops it in that component's unique history, and the mixture does not say which component is actual. The witness is a different object. It is a valid assignment of probabilities, and the stopping, if one insists on that word, is not the stopping pattern of any mixture of rules that are consistent for every intervention.

Probabilistic validity is weaker than either of those descriptions. A valid classical process need only assign total probability one under every product of local stochastic maps. It may do so without being a mixture of process functions, and it may do so without being purifiable. The witness shows the first gap, from the hull to the purifiable processes. The unpublished note records examples of the second gap, from the purifiable processes to the merely valid ones, and those examples are not proved again here. Keeping the two gaps apart is what prevents a clean slogan. One cannot say that quantum theory cuts the classical processes back to the deterministic consistent ones. One also cannot say, on the strength of the witness alone, that every valid classical process is purifiable. The old conjecture said the first. It is false. The second remains open.

2.3 One loop, counted by hand

The condition is not vacuous, and it is not a disguised demand that the parties be ordered. The three-bit rule often called the Lugano process sends the output bits (a, b, c) to the input bits

$$(\neg b \wedge c,\ \neg c \wedge a,\ \neg a \wedge b).$$

Each input depends on the other two outputs. There is no party whose input is independent of the others, and there is no way to list the parties so that each depends only on earlier members of the list. The dependence is a cycle.

The fixed-point count can be seen in one case without any machinery. Let every party return its input unchanged. A history then has to satisfy $a = \neg b \wedge c$, $b = \neg c \wedge a$, and $c = \neg a \wedge b$. If $a = 1$, then $b = 0$ and $c = 1$, and the third equation says $c = 0$. If $a = 0$, then $b = 0$ and $c = 0$, and $(0, 0, 0)$ satisfies all three equations. Exactly one history. The same count for every other triple of local bit functions is a finite enumeration. The rule is a process function. It is not an acyclic circuit rewritten in a dramatic notation.

Two lessons should be kept separate. The example shows that universal consistency under local functions does not require a party who is first. It does not show that a laboratory on a closed curve in a relativistic spacetime can implement the rule. Adlam (2023) is explicit that "possible," in this classical construction, means logically possible relative to the constraint, and that it is not known whether the dynamics can be implemented in the actual world. That caution is a premise of this article, not a concession added at the end.

3 Fixed causal order is a further demand

A fixed causal order, as the phrase is used here, is a strict partial order of the parties such that signalling, if it occurs, respects the order. A party's output may influence a later party's input. It does not influence an earlier party's input, and the order itself is not a function of the parties' choices. Acyclic deterministic circuits meet this demand, and they are process functions. They are not all of the process functions. The Lugano rule is the witness already on the table. Cyclic dependence is incompatible with a strict partial order of signalling, and the fixed-point condition holds anyway.

Adlam (2023) draws a related distinction for a different purpose. She argues that the process-matrix programme has identified a notion of causal order that matters in fundamental physics, and that this notion is weaker than ordinary causation because it does not require asymmetry. Causal order, on her account, is a matter of whether the possibilities of signalling can be arranged in a strict partial order, or in a bi-order whose two directions are the two ways of reading the same undirected constraint. Ordinary causation adds a preferred direction. She also notes that the consistency conditions built into the formalism do not by themselves force every admitted process to be causally ordered. Non-causal processes can obey the conditions that keep local operations from generating contradictions. Something further would be needed to explain why the regimes we have probed look ordered.

The present argument uses that observation and then separates two claims that her discussion leaves close together. The first claim is hers: consistency under local operations does not entail a fixed causal order. The Lugano process is a classical, deterministic, fully checked instance. The second claim is not a reading of her paper. It is the reversible-extension theorem of Section 4. A process can fail to be ordered, meet the fixed-point condition, and still possess a unitary realization of a stated kind. The failure of order is not repaired by the passage to unitaries. Readers who treat "consistent," "reversible," and "causally ordered" as three names for one virtue of a physical process are running the three together.

The two-party case shows where the split cannot appear. For two parties, the valid classical processes are mixtures of causally ordered process functions. The unpublished note records this as a standard fact and re-proves it for arbitrary finite alphabets. Adlam (2023) makes the corresponding point in the language of the classical theory: on a closed curve with one or two laboratories, the only process functions are trivial. The interesting coexistence of universal consistency and cyclic dependence begins at three parties. That is also the smallest setting in which a process function can violate a causal inequality. Violating a causal inequality and lying outside the convex hull of the process functions are different facts. The Lugano process does the first and not the second. It is a process function, so it lies in the hull, and its dependence is cyclic. The witness of Section 6 does the second. It is not offered as a new causal inequality.

There is a temptation to hear "no fixed causal order" as "time travel has been shown." The temptation should be resisted for a reason that is already internal to the definitions. The absence of a strict partial order is a fact about functional dependence among alphabet symbols. It is not a fact about the light-cone structure of a spacetime, and it is not a fact about a traveler whose personal time disagrees with external time. Lewis's question, discussed in Section 5, is a question about that disagreement. A cyclic look-up table does not answer it.

4 Reversible realization

4.1 What the extension does

Reversibility is easy to overstate because a weak form of it is automatic. Add a source, whose output alphabet is the alphabet of input tuples, and a sink, whose input alphabet is the alphabet of output tuples. For an arbitrary rule w, not only for a process function, there is a permutation U of the joint basis that writes $w(o)$ shifted by the source symbol into the input registers and writes the output string o into the sink. A permutation of a finite basis is unitary. Baumeler et al. (2019) already use a reversible realization of this broad kind: deterministic dynamics compatible with arbitrary local operations can be realized by reversible interactions. The permutation U is not, by itself, the quantum question.

The quantum question is what happens when the parties do not merely apply classical functions. Each party may insert a unitary from the input system, tensored with a private finite ancilla, to the output system, tensored with a private finite ancilla, with the dimensions matched so that a unitary between them can exist. Wiring those unitaries into U defines an operator M from the source and the incoming ancillas to the sink and the outgoing ancillas. The demand is that M be unitary for every such choice. Araújo et al. (2017) call a process pure when it meets this demand. Purity is stronger than the existence of one unitary dilation. A single favorable choice of local unitaries does not establish it.

The first theorem is that every process function meets the demand. The reason is not that U is a permutation. The permutation exists for rules that are not process functions, and those rules fail the demand. The reason is a combinatorial feature of process functions that the appendix calls exclusivity. If two output strings differ, some party has a different output bit, or more generally a different output symbol, and the same input symbol, at those two strings. When the Gram matrix of M is written as a product over parties, that party contributes a factor of zero. Rows of M that ought to be orthogonal are orthogonal. The dimension count makes M square, so M is unitary. The ancillas may be entangled with each other and with the source in the incoming state. Entanglement is a feature of the state, not an extra hypothesis on w.

The same exclusivity fact explains why a process function need not be a bijection on its own alphabet. The Lugano rule is two-to-one on some slices. Its classical inverse is not a function of the output alphabet alone. The sink holds the output string, and the source, read backwards, holds the difference between the input and $w(o)$. Reversibility is restored by the source and the sink. It is not restored by forcing the classical rule to be a permutation. A reader who thinks that a consistent loop is reversible only when the looped map is a one-to-one function of the past has a different model in mind.

4.2 What unitarity adds, and what the converse restricts

The converse is easy to state and easy to over-read. Take any rule between finite alphabets, process function or not. The closed loop M is unitary for every choice of local unitaries and finite ancillas if and only if the rule is a process function. One direction is the theorem just described. The other direction is a counting argument. If some local functions have N fixed points and N is not one, there is a choice of local unitaries, a copy of those functions written as a permutation of a basis with an ancilla, for which a column of M has squared norm N. An operator with a column of the wrong norm is not unitary. Zero fixed points and two or more fixed points are both detected. One failing intervention is enough to reject the rule. The forward direction requires every intervention to pass.

The restriction is the whole content of the caution. The converse characterizes process functions inside this family of tests: the canonical extension, every local unitary, every finite ancilla. It does not say that an arbitrary operator which happens to be unitary is a process function, and it does not say that consistency and unitarity are the same idea. Column norms are not a substitute for the orthogonality

argument. A rule can have the right column norm for some insertions and fail orthogonality for others. The two directions use different facts. Orthogonality uses exclusivity. The column norm uses a specially chosen unitary as a counter. Collapsing both directions into the slogan "unitarity if and only if consistency" drops the restriction that makes the slogan true.

What the equivalence does say is still philosophically useful. Inside the process theory, for finite alphabets, one does not need an extra quantum filter, imposed on top of the fixed-point condition, in order to keep this canonical extension unitary. If the classical rule already has exactly one history for every local function, the local unitaries do not open a new inconsistency. If some local function has the wrong number of histories, there is a unitary insertion that exhibits the failure as a failure of unitarity. Quantum theory, in this limited sense, is not a further chaperone. The chaperone, such as it is, is the fixed-point condition itself.

Validity under general local instruments, including instruments that are not unitary, follows by a dilation. A local completely positive trace-preserving map on a finite system can be written as a unitary on a larger finite system, followed by discarding an environment. The purity hypothesis supplies a unitary on the enlarged loop. Discarding the environments returns a trace-preserving completely positive map. The step uses finite dimension. It is recorded in the appendix as a bookkeeping lemma, not as a new physical principle. It says that the same rules which survive every unitary insertion induce well-defined probabilities for every local instrument. It does not say that every mathematically valid classical process arises in this way.

4.3 Why the extension is not a spacetime license

The reversible extension is an operator on finite-dimensional Hilbert spaces associated with alphabets. It is not a solution of Einstein's equation, and it is not a local Hamiltonian on a wormhole. Post-selecting a circuit can reproduce a list of events conditional on success. That is a filter on samples. It is not this dilation. Vilasini and Renner (2024) prove no-go results about embedding cyclic information-theoretic structures in acyclic spacetimes. The lifting theorem is not an exception to those results, because the lifting theorem is not an embedding theorem. A reader who wants a closed timelike curve in a relativistic spacetime has asked a question this construction does not address.

The same restraint applies to freedom. Baumeler et al. (2019) show that consistency with arbitrary local operations is compatible with nontrivial three-party time travel in their classical model, and their title names freedom of choice. The freedom in the theorem is the modelling freedom to insert any local operation of the relevant type. The parties are not derived from a libertarian theory of agency, and the proof does not show that an agent in the actual world can choose a bit flip on a closed curve. Adlam makes the parallel point when she says that the classical constructions establish logical possibility relative to the constraint. Treating the quantifier over local functions as a metaphysical power is a change of subject.

Lewis's own caution points the same way, from the other direction. He defends the consistency of a certain kind of fiction, and he remarks that a parallel defense might be tried for physical hypotheses such as circular time, but he does not give that defense (Lewis, 1976). A unitary extension of a process function is not the missing defense. It is a theorem about alphabets. The fiction can be consistent. The alphabet model can be unitary. Neither fact puts a traveler on a curve.

5 Logic's bidding

5.1 Lewis's two contexts

Lewis's treatment of Tim is the standard solution that later authors refine. Tim can kill Grandfather, in one context, because he has the rifle, the skill, the line of fire, and no chaperone. Tim cannot kill Grandfather, in another context, because Grandfather lived, and killing him is not compossible with that fact. Lewis's sentence is that "can" is equivocal: to say that something can happen is to say that its happening is compossible with certain facts, and the context chooses the facts (Lewis, 1976). The forces of logic, he insists, will not stay Tim's hand. The failure, when failure is what consistency requires, has an ordinary cause. A noise, a slip, a change of mind. The same ordinary cause could have saved Grandfather's partner from a different sniper who is not a time traveler.

Loewenstein (2022) reconstructs this reply, together with the replies of Ismael (2003) and Sider (2002), as a selection effect. Once the story says that the traveler is there to make the attempt, the story has already selected the cases in which the earlier self survived. Comparing a thousand failures with a thousand ordinary coincidences then looks like a confusion about the reference class. Loewenstein quotes Ismael to the effect that a description of a self-defeating chain smuggles in the last event under the guise of the first, so that only unsuccessful cases are considered. Sider is cited by Loewenstein as arguing similarly. This article has not re-read Sider's argument and does not attribute to him any thesis beyond that citation. The Lewisian core, which has been re-read, is the contextual compossibility claim and the refusal of a logical chaperone.

Lewis's comparison with the other sniper is the part of the argument that makes the refusal concrete. Tom is not a time traveler. He has the same rifle, the same intent, and a different victim, and there is no doubt, by ordinary standards, that Tom can kill. If Tim fails for a commonplace reason, Tom could have failed for that reason as well. The wider set of facts, including the fact that the victim lived, is compossible with neither success. In Tom's case we do not reach for a chaperone, because we see that the facts which rule out success are facts about what happens after the relevant moment of ability, or facts that merely restate the failure. Lewis makes the same point against a fatalist use of foreknowledge. A truth about what one will not do, sitting in the past only because it is a truth, is not the sort of fact that removes an ability. The fact that counts, when we say someone cannot do a thing, is a fact of the kind we already treat as limiting ability in ordinary cases: the state of the weapon, the training, the obstruction. A future truth that redescribes the outcome does not join that list.

This is why a process function should not be advertised as Lewis's chaperone, and also why it should not be advertised as his whole solution. The fixed-point condition is a fact about a rule, in force for every intervention, not a truth that redescribes one outcome after the outcome has been put into the story. In that respect it is closer to a standing constraint on what counts as an admitted dynamics than to the fatalist's idle truth. It is not, however, a fact of Tom's kind. Tom's ability is limited, when it is limited, by the arrangement of a scene. The process function does not describe a scene. It describes which look-up tables are allowed to connect the laboratories. One can grant Lewis the contextual analysis of "can" inside a story, and grant him that logic does not stay a hand, without treating a Boolean rule on three bits as that story. The rule is evidence that a universal consistency condition can be stated without a personified force. It is not evidence that Tim's failure has an ordinary cause.

Loewenstein accepts that the inexplicability would vanish if the traveler can succeed relative to the facts before the attempt and cannot succeed relative to all the facts. She argues that the time-travel case does not earn the first of those two permissions in the way the telephone case does. Her conclusion is that something would still have to do logic's bidding. The banana peel does not merely happen to be there. It has to be there if the attempt is made. If one refuses a picture in which logic enforces itself through a deviant causation, one should, she argues, deny backwards time travel in a non-branching

universe.

5.2 What the quantifier supplies

A process function is a precise version of a demand Loewenstein puts in metaphysical language. The demand is not that one named attempt fails for one named reason. The demand is that every local intervention has exactly one history. In the alphabet model, that demand is not enforced by an extra object standing beside the parties. It is the admission condition on the global rule. Rules that fail it, including the identity channel around one laboratory, are not process functions. Rules that meet it may still be cyclic. The Lugano process meets it and is cyclic. The unitary theorem adds that, for finite alphabets, meeting it is exactly the condition under which the canonical extension survives every local unitary.

This is a better description of the constraint than the chaperone image, and it agrees with Lewis that one should not invent a personified force whose job is to spoil the shot. The constraint is structural. It is also, in one respect, stronger than the constraint that makes a single grandfather story consistent. The story quantifies over one intention. The process function quantifies over all functions on the alphabets. A rule that is consistent for the intentions we find narratively natural, and inconsistent for a bit flip nobody in the story bothers to attempt, is not a process function. The universality is what connects the model to Adlam's point that the parties are not to be forbidden the locally permitted operations. If the bit flip is a locally permitted operation, the rule must be able to live with it.

The model also separates determination from order. "The local choice determines a unique history" does not mean "the history is produced by a cause that is earlier in a fixed order." In the Lugano example the unique history for the identity intervention is $(0, 0, 0)$, and the dependence among the bits is cyclic. Determination, in the fixed-point sense, is a fact about the number of solutions. Order is a fact about whether those solutions can be stratified. Loewenstein's question is often heard as a question about determination: why is the outcome the one that logic needs? The process-function condition answers a formal version of that question by building the count into the rule. It does not answer it by exhibiting an earlier event that makes the later event necessary. Readers who use "determine" for both the count and the stratified production will take the theorem to have settled more than it has.

5.3 Why this does not refute the inexplicability objection

The structural answer is open to a charge of relabelling, and the charge should be granted as far as it goes. Suppose someone asks why, in a world of travelers and banana peels, the peel is there whenever it is needed to prevent a contradiction. The reply that "the dynamics are a process function" names a class of look-up tables. It does not say why a world contains a look-up table of that class rather than the identity channel. Adlam already notes a form of this worry. The framework distinguishes local operations, which may be chosen freely, from the dynamics between the laboratories, which may not. One might think that the distinction is too strong, because dynamics look like chains of local operations. If that complaint is right, the process function has restated the exclusion of the identity channel rather than explained it.

Nothing in the lifting theorem repairs the complaint. The theorem says that if the table is a process function, the canonical quantum extension is unitary under local unitaries, and that if the extension meets that standard, the table is a process function. It moves the constraint from a classical count to a unitary count without supplying a mechanism that places banana peels. Loewenstein asks for a kind of modal robustness: the preventing condition must be there, not merely happen to be there. A process function provides robustness inside the model, because the unique-history clause is quantified over all interventions, not discovered after one intervention has been chosen. That is robustness of a rule

under substitution of local functions. It is not robustness of a cause across the nearest worlds in which a traveler forms an intention. The second robustness is Loewenstein's topic. The first is the topic of the theorem. Using the theorem as a refutation of her objection would require them to be the same topic.

A similar limit applies to Lewis. His claim is that a consistent time-travel story is possible, and that the relevant "can" is contextual. A process function is one way to write a consistent story in a very restricted language. Its existence supports the modest claim that consistency, under a stated quantifier, is not an empty demand. Lewis did not need a three-bit Boolean rule to make that claim, and the rule does not vindicate his account of personal time, of stages, or of counterfactuals. Loewenstein's disagreement with Lewis is about whether the contextual dissolution leaves a real inexplicability. The alphabet model can be described without taking a side in that disagreement. It can say: here is a constraint that is not a chaperone, that is compatible with cyclic dependence, and that is necessary and sufficient for a stated unitary extension. It cannot say: therefore the banana peel is no longer mysterious, or therefore time travel is possible.

Loewenstein (2022) also rejects an appeal, which she attributes to Baron and Colyvan (2016), to the joint work of a program explanation and a process explanation. On her telling, the program explanation says why the traveler must fail if success would be contradictory, and the process explanation says which ordinary event in fact produced the failure, but neither explains the modal fact that a preventing condition had to be in place. This article has not re-examined Baron and Colyvan's text. It uses Loewenstein's distinction only as a warning about the process function. The fixed-point clause resembles a program constraint: it says what must be true of any admitted rule. The particular history of a particular intervention resembles a process explanation: it says which output tuple occurs. Neither is the spacetime mechanism Loewenstein is asking for. The resemblance is a reason to keep the philosophy modest, not a reason to conscript her opponents.

6 Purification and the chosen basis

6.1 Mixtures

The inclusion that does hold is straightforward. A mixture of process functions is purifiable. Prepare a control register in a superposition whose squared amplitudes are the weights, apply the reversible extension of the selected process function, and route the control to the sink. The closed loop is block diagonal in the control basis, with unitary blocks, so it is unitary. Discard the sink and read the computational basis. The classical statistics are the mixture. Purity, in Araújo and coauthors' sense, is therefore compatible with ignorance over deterministic consistent rules. If that were the end of the story, purification would be a way of saying that the classical process was a mixture of process functions and that the mixture had been written with coherent phases.

The converse was a natural conjecture. It is false. The unpublished note conjectured that the purifiable classical processes were exactly the mixtures of process functions. The conjecture is refuted by one three-party binary example. It is not refuted in the sense that every neighboring conjecture is false. The shape of the set of purifiable processes remains open: whether it is a polytope, which further valid processes it contains, and which inequalities that hold on the hull survive for every purifiable process. The note records exact certificates that some valid processes are not purifiable. Those certificates are not re-proved here. If they are granted, the purifiable processes sit strictly between the hull and the set of all valid classical processes. If they are not granted, the one strict inclusion still stands. The hull is a proper subset of the purifiable processes.

6.2 What the witness excludes

The witness is a classical conditional distribution on three binary parties. It is obtained from an explicit pure process by preparing a stated source state and reading the computational basis. The source state is an equal superposition of two computational-basis strings, with a relative phase of $-\pi/4$. The board that defines the process is a family of three-qubit unitaries, one for each output string, each diagonal in the Hadamard basis. The induced distribution p is nonnegative and normalized under every product of local stochastic maps, by a finite exact check. It is therefore a valid classical process.

A linear functional L separates it from the hull. On the witness,

$$L(p) = \frac{1}{4} - \frac{3}{16}\sqrt{2} < 0.$$

On each of the 744 process functions of three binary parties, L is nonnegative, and the bound is tight on many of them. Nonnegativity on the vertices passes to the convex hull. So p is not a mixture of process functions. The number 744 is the exhaustive count of the fixed-point definition on three bits, not an estimate. The value of $L(p)$ is exact. A floating-point distance from p to the hull is a check, not the proof.

What this excludes is a specific reading of the probabilities. One cannot say that p, in the computational basis, is the record of ignorance about which process function was in force. Every mixture would have given L a nonnegative value. The witness does not exclude other readings. It does not say that the probabilities are produced by a chance process in spacetime. It does not say that no other hidden variable, of a different type, could be invented. It says that the hidden variable cannot be a process function, or a mixture of them, if the observed conditional distribution is p.

The board is not a new quantum process in every local frame. The unpublished note identifies it, up to a local frame, as a time-reversed variant of a Lugano purification in the family studied by Guérin and Brukner (2018), with one party's conditional bit flip replaced by a square root of that flip. The identification is a claim of the note about a known family. It is not a claim to have discovered the board. Read in the computational basis, the induced classical process lies outside the hull. Read in another basis, a pure process of this family can induce a mixture of process functions. The separation is a statement about a named frame. Frame dependence is not a defect that appears after the theorem. It is part of what the theorem says. A classical distribution is induced by a choice of measurement, and the choice has to be stated before the distribution is an object that can lie inside or outside a classical convex hull.

Two further boundaries keep the witness from being asked to do too much. The gap requires three parties. The two-party case collapses, as Section 3 recorded. And the witness is not a device-independent causal inequality. Whether this particular distribution violates such an inequality is a different calculation. The article does not make it. Causal-inequality violation remains the phenomenon illustrated by process functions such as Lugano, which lie inside the hull.

6.3 Adlam's order, and what purity does not restore

Adlam's central thesis is that fundamental physics can contain causal order without containing causation in the ordinary asymmetric sense. The process-matrix notion, she argues, is a modal constraint on signalling. Choose a direction through the constraint, by a perspective or by an interventionist practice, and ordinary causal talk can be recovered in restricted domains. Do not choose a direction, and there is still a fact of the matter about whether any direction is available. An underlying process that admits no ordered description is not saved by redescribing the inputs and outputs. She is also careful about the quantum switch and about no-signalling correlations. Those discussions are not the target here. What matters is her use of consistency and of purification.

The partial character of the consistency explanation is the part of her discussion that the witness touches. She argues that the conditions built into the process formalism explain a good deal of the prominence of causal order. Signalling processes can be arranged into contradictions of the one-laboratory kind, so a constraint that forbids those contradictions pushes signalling processes toward an order. No-signalling processes are not under the same pressure, because they cannot be wired into that contradiction, and she treats this as a reason not to expect a causal-order analysis of correlations that do not signal. The explanation stops short of the actual world, on her own account, because the same formalism still contains non-causal processes that obey the consistency conditions. Consistency is why a signalling contradiction is excluded. It is not why every admitted process is ordered. Purification is one of the further postulates she canvasses, following Araújo and coauthors, precisely because it rules out a large class of non-causal processes and not all of them.

That is the right place to locate the three-party witness, and the wrong place to announce a refutation. If purification returned every classical record to the convex hull of the process functions, it would be a device that converts the surviving pure processes back into mixtures of deterministic consistent rules, cyclic rules included. The witness says that the device does not work that way. A pure process can induce a classical distribution that is valid, hence consistent in the probabilistic sense of total probability one, and that is not a mixture of the deterministic rules. The further postulate does real work. It is not the work of restoring the hull. Adlam's question, why the probed regimes look ordered even though consistency permits non-ordered processes, is therefore not answered by identifying purity with membership in the hull. The question remains a question about what, beyond the fixed-point condition and beyond purity, would exclude the processes we do not see.

On consistency, the present results agree with her and sharpen the classical side. The conditions that keep local operations from producing contradictions do not force a fixed causal order. The Lugano process is a finite object for which both halves can be checked by hand: the order fails, and the fixed-point count holds. The unitary theorem adds a sentence she does not need for her conclusion and that should not be borrowed as if it were her conclusion. The same unordered process function has a reversible extension that remains unitary under every local unitary. Failure of order survives the passage to the pure process. Reversibility of this kind does not put the asymmetry of ordinary causation back in, and it does not put a strict partial order back in either.

On purification, a misreading is available and should be blocked. Araújo et al. (2017) propose that physical processes are those that can be obtained from pure processes. Adlam reports the proposal as ruling out a large class of non-causal processes, not all of them, and she treats it as one of the further ingredients that might explain why the probed world looks causally ordered. In an appendix she notes that pure processes induce unitary, hence reversible, transformations when the laboratories apply unitaries, and she says it appears that causal processes can be derived from time-reversal-invariant underlying physics. She then conjectures that causal processes are reversible under a suitable operational time reversal, or can be extended by an ancilla to processes that are. She states that her argument does not depend on the conjecture. The conjecture is about causal processes and about operational reversibility. It is not the conjecture that every classical distribution induced by a pure process is a mixture of process functions.

The witness bears on the second conjecture, which was the unpublished note's, and not on the first, which is hers. A pure process can induce a classical distribution outside the hull. Purity does not mean "mixture of deterministic consistent histories," once a basis has been fixed and the distribution is the computational-basis one. That blocks a picture in which the purification postulate quietly restores the deterministic consistent processes as the only classical records. It does not block Adlam's claim that causal order is a weak symmetric structure. It does not refute her conjecture about the operational reversibility of processes that are causally ordered. A causally ordered process and a computational-basis marginal of a cyclic pure process are different objects. A theorem about the second is not a theorem

about the first.

There is a related moral about the phrase "time-symmetric underlying physics." The inverse of the reversible extension exchanges the roles of source and sink. That exchange is a symmetry of an operator. It is a candidate for the sort of underlying reversibility Adlam is interested in, and it can be pointed out without claiming that it meets her operational definition of time reversal. The mathematics forces a further discipline. Before the operator says anything about classical statistics, a frame and a source state have to be named. In one frame the statistics may be a mixture of process functions. In another they may lie outside the hull. Time symmetry of the board is not a verdict on the classical process. Readers who move from "the pure process is reversible" to "the classical loop was a consistent deterministic history, up to ignorance" are using a step the witness removes.

The basis is not an embarrassment peculiar to this example. It is the point at which a quantum process becomes a classical process. Adlam distinguishes an input-output description, in which some variables are treated as inputs and others as outputs, from the underlying process that can be cut in more than one way. The computational-basis reading is one cut. A Hadamard-basis reading is another. The hull is a set of classical processes, so membership in the hull is a question asked after the cut. The theorem that p lies outside the hull is a theorem about that question. It is not a theorem about a basis-independent classical reality hiding under the board.

7 Objections

7.1 There is no philosophical payoff

The first objection says that the results are algebra, and that the philosophy was already available without them. Lewis already showed that consistency is not impossibility. Adlam already showed that consistency does not force causal order. Araújo and coauthors already defined purity. A three-bit calculation does not advance those disputes.

The objection is right about the ownership of the disputes and wrong about the payoff. The payoff is a set of separations that the disputes need once they are stated in a common vocabulary. Consistency under all deterministic local functions, membership in a fixed causal order, reversible realization of the canonical extension, and being a mixture of process functions are four conditions. The Lugano process meets the first, fails the second, and meets the third. The witness meets a pure-process condition, induces a valid classical process, and fails the fourth, in a named basis. Acyclic circuits meet the first and the second. The identity channel around one laboratory fails the first. These are not paraphrases of one distinction. A philosophical argument that treats them as paraphrases will mislocate what has been shown. In particular, it will treat a unitary extension as evidence of time travel, or a pure process as evidence that the classical record was a consistent deterministic loop. Both inferences are invalid, and the invalidity is the philosophical result.

The result is modest on purpose. It does not offer a theory of causation, a theory of agency, or a theory of explanation. It says which inferences among these four conditions are safe inside a stated formalism, and which inferences are equivocations. That is a contribution to the disputes only because the disputes have been using the conditions as if the inferences were safe.

7.2 The constraint is trivial because it is built in

The second objection says that a process function explains nothing, because the fixed-point clause is the admission condition. Anything that is admitted will, trivially, have exactly one history. The appearance of explanation is the appearance of a filter that was installed by definition.

The objection is the relabelling charge of Section 5.3, and the right reply is to say what the definition is doing and what it is not doing. It is not predicting that the actual world contains process functions. It is sorting rules. The sorting is not empty. Some familiar rules fail it, and some cyclic rules pass it. The unitary theorem is not part of the definition. One could have defined process functions by the fixed-point clause and then discovered that their canonical extensions fail to be unitary for some local unitary. That discovery would have been the discovery of an extra filter. The proof says that there is no such extra filter, and the converse says that the filter one does have is detected by a unitary test. A definition that merely repeated the fixed-point clause in the language of unitarity would not need the exclusivity argument or the column-norm argument. Those arguments are what connect two descriptions that were introduced apart from each other.

The objection would be decisive if the article claimed to have explained why nature uses process functions. It does not. Adlam's further question, why the regimes we probe look causally ordered even though consistency permits non-ordered processes, is still open. Purification was one proposed answer, in the form of a postulate that excludes some non-causal processes (Araújo et al., 2017; Adlam, 2023). The witness limits that answer. Purity excludes whatever it excludes, and it does not exclude every classical process outside the hull. A postulate of purity is not a postulate that the classical record is a mixture of process functions.

7.3 The basis spoils the separation

The third objection says that a frame-relative separation is not a separation. If the same board induces a mixture in another basis, then there is a reading on which the process is a mixture of process functions, and the computational-basis calculation is an artefact of a bad cut.

The objection identifies a real fact and draws the wrong conclusion from it. The real fact is that the hull is a set of classical processes, and a classical process is induced only after a frame is chosen. The wrong conclusion is that a negative result in one frame is empty unless it holds in every frame. Classical probabilities in a laboratory are probabilities in the frame the laboratory measured. If those probabilities are the witness, they are not the probabilities of a mixture of process functions, even if some other experiment, with the local frames rotated, would have produced a mixture. Observer-dependent locality, in the sense studied by Guérin and Brukner (2018), is a reason to expect this kind of sensitivity. It is not a reason to ignore the distribution that a stated measurement produces.

The objection would succeed against a different claim, which this article does not make. The claim would be that the board fails to purify any process function in every frame, or that the board is a new causal structure invisible from every description. The note denies both. The board is known, and the separation is the separation of one induced classical process from one classical convex set. A philosophy that wants a basis-independent verdict has to ask a basis-independent question. "Is this classical distribution a mixture of process functions?" is not that question. It only looks basis-independent when the basis has been left implicit.

7.4 Infinite cases are artefacts

The fourth objection says that the infinite-party remarks are set theory intruding on a physical question, and that they should not be used to draw a moral about consistency.

The objection is right about the use this article makes of them. They are not premises of the finite theorems. They are a boundary. With infinitely many parties, the exclusivity property still yields at most one fixed point. If each local map depends continuously on the outputs, it yields exactly one. Without a continuity or definability restriction, there is a set-theoretic example of an exclusive self-map of the space of infinite binary sequences with no fixed point at all. The corresponding infinite-party rule is not

a process function, because existence fails. The general question whether every exclusive map of Baire class one on that space has a fixed point is open. It is not answered by the example, and the example is not presented as an exclusive map of Baire class one.

The moral, which is modest, is that a pairwise condition does not by itself guarantee a global history once the parties are infinite. That moral is a reason to keep the finite hypothesis visible in the main theorems. It is not a paradox of time travel, and it is not a resolution of any infinite-regress puzzle in metaphysics. Those puzzles are not analyzed here. Using the set-theoretic example as a counterexample to a philosophical view of regress would repeat the mistake this article is written to avoid: taking a theorem beyond the question it answers.

What remains open inside the finite theory is enough to discourage a larger moral. The set of purifiable classical processes is not characterized. Continuous alphabets are a different problem, because a single history is typically a null set. The passage from finite ancillas to arbitrary Hilbert spaces is part of the unpublished note for the witness and is not re-proved here. None of these open ends turns the finite separations into conditional claims. The Lugano process is a process function. Its extension is unitary under local unitaries. The witness lies outside the hull. Those three sentences do not depend on the infinite theory or on the unproved certificates.

8 Conclusion

Universal consistency under local functions of a stated type can hold when no fixed causal order holds. For finite alphabets, that consistency condition is necessary and sufficient for the canonical reversible extension to remain unitary under every local unitary and every finite ancilla. The necessity and the sufficiency are theorems about that extension. They are not a definition of unitarity, and they are not a theorem about spacetime. Mixtures of the consistent deterministic rules are purifiable. The converse fails for an explicit three-party distribution in the computational basis, induced by a known board. Purity is a weaker classical constraint than the fixed-point condition, once the basis is fixed.

The philosophical use of these facts is negative and specific. They block the inference from a consistent cyclic rule to a fixed causal order. They block the inference from a reversible extension of such a rule to a physical or metaphysical time-travel claim. They block the inference from purity to the conclusion that the induced classical process is a mixture of process functions. They do not block Lewis's claim that a consistent time-travel story is possible, Loewenstein's claim that a remaining inexplicability attaches to backwards time travel in a non-branching world, or Adlam's claim that causal order is a weak symmetric structure that ordinary causation refines. Those claims are about topics the theorems do not settle. The service of the theorems is to keep the topics from being settled by accident, through a vocabulary in which consistency, order, reversibility, and purification share a name.

9 Assistance and declarations

Preparation of this draft, including the organization of the argument and the wording of the appendix, was assisted by Grok Bot, a Cursor cloud agent. The assistance was not limited to copy editing. It included generative drafting. The assistant is not an author. Large language models do not meet authorship criteria. The underlying model and provider, beyond the description just given, are not recorded in this draft. There must be human accountability for a final text. Author verification of this draft is pending, and no such verification is claimed here.

The mathematical statements in the appendix are also set out, for a different readership and with different aims, in an unpublished physics-style manuscript by Arrowood. This article's contribution is the philosophical argument. Neither manuscript is asserted to be unique, and neither has been submitted.

Funding has not been confirmed. No funding statement should be inferred. Competing interests have not been confirmed. No declaration of the absence of a conflict should be inferred. Ethics approval and consent do not apply: the work is theoretical. Access to supplementary calculations under a masked author name has not been arranged. The appendix is the self-contained mathematical content offered with this draft.

A Definitions and short arguments

The unpublished note contains longer proofs, tables, and machine-checked certificates. This appendix states the objects the article uses and the short arguments that connect them. It is not a second paper.

A.1 Process functions and the extension

Let each party k have finite nonempty alphabets I_k and O_k, and write I and O for the products. For the permutation form of the extension, identify I_k with the cyclic group of the same cardinality. A local intervention is a tuple of functions $f_k : I_k \to O_k$. A rule $w = (w_k) : O \to I$ is a process function when, for every intervention, the self-map

$$o \longmapsto \big(f_k(w_k(o))\big)_k$$

of O has exactly one fixed point.

The reversible extension uses a source system P with basis $\{|e\rangle\}$ indexed by I and a sink system F with basis indexed by O. For every rule w,

$$U|o\rangle_O|e\rangle_P = |w(o) + e\rangle_I|o\rangle_F$$

is a permutation of a finite basis, hence unitary. Given unitaries V_k from $I_k \otimes X_k$ to $O_k \otimes Y_k$, with finite ancillas and $|I_k|\,|X_k| = |O_k|\,|Y_k|$, the closed loop is the operator M from $P \otimes X$ to $F \otimes Y$ with

$$M\big[(s, y), (e, x)\big] = \prod_k V_k\big[(s_k, y_k), (w_k(s) + e_k, x_k)\big].$$

A process is pure, in the sense used in the article, when M is unitary for every such choice (Araújo et al., 2017). A classical process is purifiable when it arises from a pure process by a source state, a stated orthonormal frame, and the discard of the sink. Unless a frame is named, the frame is the computational basis.

A.2 Exclusivity and the lift

If w is a process function and $s \neq s'$ in O, some party k has $s_k \neq s'_k$ and $w_k(s) = w_k(s')$. Otherwise $w_k(s) = w_k(s')$ would imply $s_k = s'_k$ for every k. Define f_k to take the value s_k at $w_k(s)$ and the value s'_k at $w_k(s')$, and define it arbitrarily elsewhere. The two clauses agree where the inputs agree, and s and s' are distinct fixed points of one intervention.

For a process function, the Gram matrix of M factorizes over parties. If $w_k(s) = w_k(s')$, the change of variables $i = w_k(s) + e_k$ makes the k-th factor the inner product of two rows of V_k. If $s = s'$, every factor is of this kind and the product is the inner product of the ancilla labels. If $s \neq s'$, exclusivity supplies a party whose factor vanishes. Thus $MM^\dagger$ is the identity. The dimension hypothesis makes M square, so M is unitary. This is the lifting theorem. It quantifies over all finite ancilla dimensions and all unitaries of those dimensions.

Conversely, suppose an intervention f has $N \neq 1$ fixed points. Take $X_k = O_k$ and $Y_k = I_k$, and

$$V_k|i, x\rangle = |f_k(i) + x,\ i\rangle.$$

This is a permutation, hence unitary. The squared norm of the column $(0, 0)$ of M equals N. So M is not an isometry. Combined with the lifting theorem, the closed loop is unitary for every local unitary and every finite ancilla if and only if w is a process function. The converse uses one specially chosen family of unitaries. It does not characterize unitarity in any wider sense.

A local instrument that is not unitary dilates, on a finite system, to a unitary on a larger finite system followed by a partial trace. Apply the lifting theorem on the enlarged loop and discard the environments. The result is a trace-preserving completely positive map. The finite-dimensional padding is essential to the form of the argument given here.

A.3 The witness

Mixtures of process functions are purifiable by a control register that selects the component and is routed to the sink. The converse fails. There is a valid classical process p on three binary parties, induced in the computational basis by a pure process with a finite source, such that

$$L(p) = \frac{1}{4} - \frac{3}{16}\sqrt{2} < 0,$$

while $L \geq 0$ on every three-bit process function and hence on their convex hull. The functional L is an integer linear combination of the conditional probabilities. The unpublished note records the coefficients and the family of Hadamard-diagonal unitaries. The source vector is

$$\phi = \frac{|011\rangle + e^{-i\pi/4}|100\rangle}{\sqrt{2}}.$$

Three checks are distinct. Validity is a finite multilinear identity. Purity is a cancellation, in an exact cyclotomic arithmetic, among the Pauli coefficients that would connect distinct output strings, together with a general lemma that ancillas do not create new failures for this shape of board. Non-membership in the hull is the sign of L. Any one of these checks can fail while the others hold. The article uses all three.

The same note identifies the board, up to a local frame, with a time-reversed variant of a known Lugano purification (Guérin and Brukner, 2018). The classical separation is the sign of L in the computational basis. It is not a claim that the board is new.

With infinitely many parties, exclusivity still implies at most one fixed point. Continuous dependence of the local maps on the outputs restores existence. There is a set-theoretic exclusive self-map of $\{0, 1\}^{\mathbb{N}}$ with no fixed point, so the corresponding rule is not a process function. Whether every exclusive Baire-class-one self-map of that space has a fixed point is open. None of these statements is a premise of the finite theorems.

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